

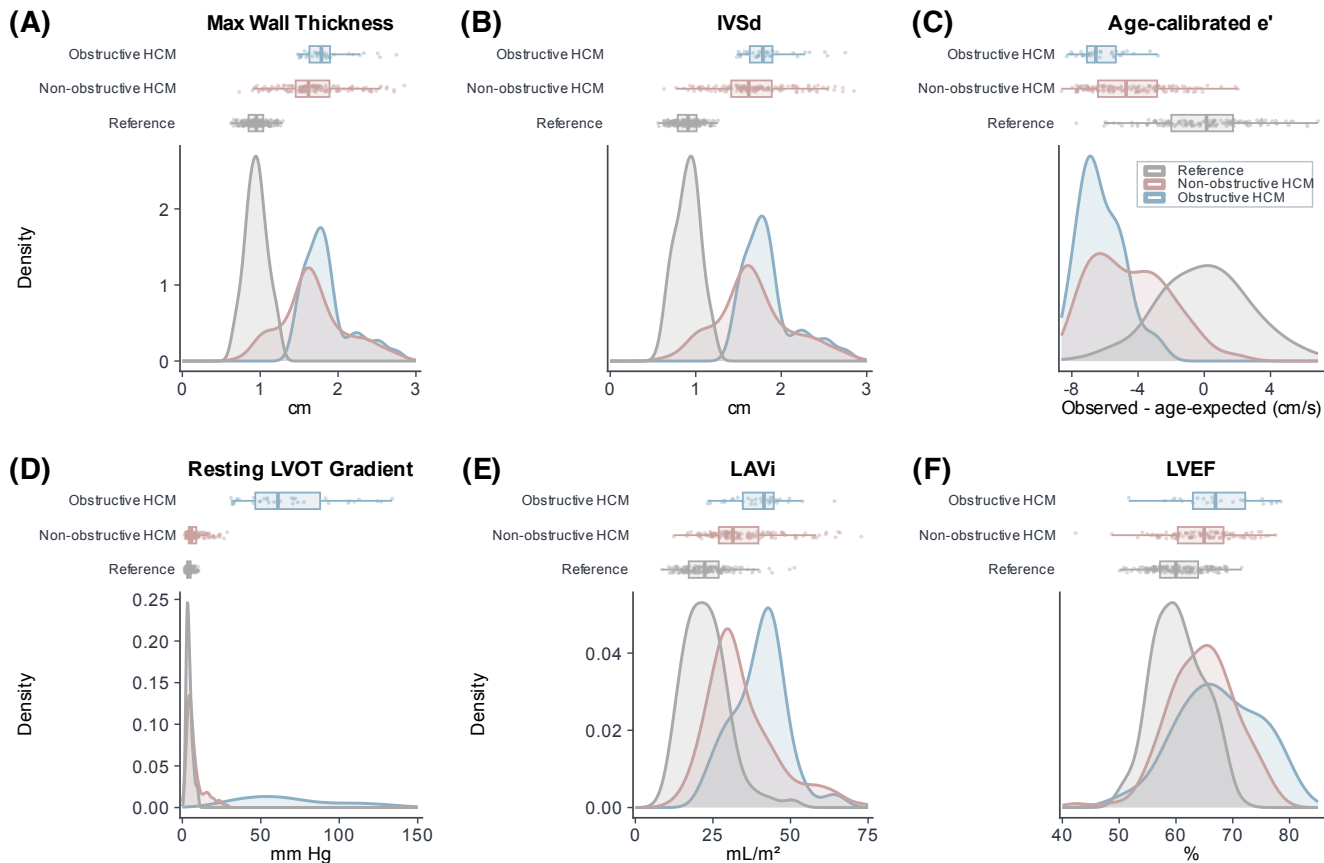


Figure 1. Matched Cohort Validation. The matched analysis includes 177 HCM patients and 177 phenotypically screened non-HCM comparators (354 total). Age-, sex-, and BMI-matched non-HCM references (gray), non-obstructive HCM (rose), and obstructive HCM (resting LVOT gradient ≥ 30 mm Hg; steel blue) are compared across six echocardiographic parameters. Each panel shows kernel density curves with a horizontally oriented boxplot and individual patient values. **(A)** Maximum wall thickness. **(B)** Interventricular septal thickness in diastole (IVSd). **(C)** Age-calibrated average e' . **(D)** Resting left ventricular outflow tract (LVOT) gradient. **(E)** Left atrial volume index (LAVi). **(F)** Left ventricular ejection fraction (LVEF).